Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Artificial Intellegence

Practical – IV

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Class:TYIT	Batch:2
Date of Assignment: 25/07/26	Date/Time of Submission:25/07/26

Aim:-Write a Python program to compute the probability that the person actually has the disease given that the test result is positive.

Scenario:
A medical test is used to detect a particular disease.
The probability that a randomly selected person has the disease is 1%.
If a person has the disease, the test returns positive with probability 99%.
If a person does not have the disease, the test still returns positive with probability 5% (false positive rate).
A person takes the test and receives a positive result.

Code:-

```
# Probability of having disease
P_D = 0.01

# Probability of positive test if disease exists
P_Pos_D = 0.99

# Probability of positive test if disease does not exist
P_Pos_NotD = 0.05

# Probability of not having disease
P_NotD = 1 - P_D

# Bayes Rule
P_D_Pos = (P_Pos_D * P_D) / ((P_Pos_D * P_D) + (P_Pos_NotD * P_NotD))

print("Probability that person actually has disease:")
print(round(P_D_Pos * 100, 2), "%")
print("Ishita Patel")
```

Output:-

```
Probability that person actually has disease:
16.67 %
Ishita Patel
```